
BAR: Budgeted Actor Refinement Decouples Bootstrap Exposure from Policy Reach

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Abstract

TD3+BC uses one actor for Bellman targets and deployment, coupling target movement to policy reach. With fixed critic and scale, its squared anchor is a conditional-Dirac Wasserstein-2 cost; ideal proximal and explicit small steps share a critic-induced gradient flow. This local picture does not predict finite-step instability. Budgeted actor refinement (BAR) splits T across K persistent actors, routes the first into TD targets, and deploys the last. Across nine D4RL tasks, fourteen budgets, and four seeds, the methods are close at small T , but at $T = 4, 7$ TD3+BC falls to 45.93, 30.36 while BAR-P4 remains at 82.06, 81.79. Over $T \geq 4$, increasing K from one to four raises the task-equal mean from 25.38 to 63.97 and cuts score- < 20 cells from 34/63 to 8/63; both disjoint seed pairs reproduce the ordering. Matched audits show that P3/P4 reduce first-branch action and target-value perturbation without typical final-policy contraction. A two-actor P3 control reaches the same seed-mean collapse count without re-centering. Thus BAR shifts the finite-step stability boundary; matched evidence is consistent with exposure-reach separation, while the chain’s benefit beyond that separation remains open.

1 INTRODUCTION

Offline reinforcement learning (RL) learns from fixed data. A central question is how far policy improvement can proceed before the critic becomes unreliable. TD3+BC anchors a deterministic actor to dataset ac-

tions while its Polyak copy drives bootstrapped targets (Fujimoto and Gu, 2021). One branch therefore sets both target movement and evaluated reach: aggressive improvement can amplify critic error, while conservative improvement can leave value unrealized.

Budgeted actor refinement (BAR) represents a nominal improvement budget with K sequential actors. The first is dataset anchored; each later actor is anchored to its stop-gradient predecessor. A target copy of the first supplies TD-target actions, while the last is deployed. BAR thus separates target construction from deployment without changing the critic update rule.

Let D_{critic} and D_{final} denote squared displacement of the critic-coupled first branch and deployed policy. A one-hop actor identifies them; BAR assigns them to μ_1 and μ_K . Throughout, *bootstrap exposure* denotes this target-action-displacement proxy, not distance to dataset support, critic error, or Bellman-target error. Separation can preserve reach at lower exposure, but also creates target-deployment mismatch. It exposes a trade-off, not independent optimality guarantees.

For deterministic actors, the squared TD3+BC anchor is a Wasserstein-2 movement cost in the Dirac-policy limit. This yields a regime hypothesis. At small realized steps, ideal JKO/proximal and explicit updates with a common frozen scale discretize the same fixed-critic flow. In the transition regime, a single large actor step can move the Bellman target beyond a critic-valid neighborhood; splitting T lets the target-routed first branch move by a smaller nominal coefficient while later branches retain deployment reach. At still larger T , projection, amortized actor fitting, critic drift, saturation, and optimizer history break the local interpretation, so depth can only shift, not remove, the failure boundary.

The phase diagram matches this pattern: no strict small-budget depth ordering, then BAR-P4 above 81 at $T = 4, 7$ as TD3+BC falls to 45.93, 30.36. Over $T \geq 4$, means rise $25.38 \rightarrow 41.20 \rightarrow 51.92 \rightarrow 63.97$ with depth and low-return counts fall $34 \rightarrow 22 \rightarrow 16 \rightarrow 8$.

Audits find lower first-branch exposure without typical final-policy contraction and critic-dependent gain reversal in failed runs. A P3 two-actor control makes separation a sufficient candidate mechanism, leaving the chain’s incremental contribution open.

Our contributions are as follows.

- We formulate BAR for TD3+BC as a re-centered actor composition that assigns target-policy displacement and final-policy reach to distinct branches.
- We connect the anchored update to an ideal Wasserstein/JKO flow, derive finite-step critic-error and target-perturbation statements, and identify where projection, ReLU boundaries, and live optimization end the small-step interpretation.
- Four-seed phase diagrams and matched audits show that multi-hop BAR becomes useful as TD3+BC exits its small-budget plateau: it reduces the target-coupled first movement, preserves deployment reach, and shifts the large-budget collapse boundary, chiefly through lower-tail rescue.

2 RELATED WORK

Behavior-regularized offline actor–critic learning. Offline actor–critic methods control extrapolation through constrained policy improvement, conservative values, or restricted Bellman-backup actions (Fujimoto et al., 2019; Kumar et al., 2019). TD3+BC maximizes a normalized critic under a squared data anchor (Fujimoto and Gu, 2021); we keep its critic and actor family fixed and change which refined branch supplies the TD target. One-step offline RL limits repeated policy optimization (Brandfonbrener et al., 2021); BAR retains iterative training but routes a budgeted actor chain.

Policy geometry and proximal refinement. Trust regions arise from information geometry (Amari and Nagaoka, 2000; Schulman et al., 2015); optimal transport instead yields Wasserstein policy flows and related continuous-control constructions (Zhang et al., 2018; Moskovitz et al., 2020; Terpin et al., 2022; Song et al., 2023). Wasserstein Policy Optimization projects such a flow into a parametric actor using value gradients (Pfau et al., 2025). We use the deterministic identity between statewise squared W_2 cost and action displacement, while separating finite-step proximal meaning from the small-step flow limit.

Target networks and actor–critic coupling.

Polyak averaging delays bootstrap feedback but normally copies the deployed branch. MCEP instead excludes a milder evaluation policy from Bellman backups (Xu et al., 2024). BAR shares this routing principle but varies a persistent, predecessor-recentered chain under fixed nominal budget; MCEP directly anchors both actors with separate strengths and is not BAR at $K = 2$. Our contribution is the depth-indexed routing operator, not a new critic objective.

3 PRELIMINARIES

We consider a discounted Markov decision process $\langle \mathcal{S}, \mathcal{A}, P, r, \gamma \rangle$ with continuous actions and a fixed offline dataset $\mathcal{D} = \{(s_i, a_i, r_i, s'_i)\}_{i=1}^N$. A clipped double critic is trained from targets

$$y_\rho = r + \gamma \min_j Q_j^-(s', \rho(s')), \quad (1)$$

where ρ is a target policy. When $\rho(s')$ leaves dataset support, critic error enters the regression target and can be amplified by later actor updates (Fujimoto et al., 2019; Kumar et al., 2019).

For a deterministic actor μ_θ , normalized TD3+BC minimizes

$$\mathcal{L}_{\text{actor}}(\theta) = -\alpha \mathbb{E}_s \bar{Q}(s, \mu_\theta(s)) + \mathbb{E}_{(s,a) \sim \mathcal{D}} \frac{\|\mu_\theta(s) - a\|_2^2}{d}, \quad (2)$$

where $\bar{Q} = \hat{Q}/C$ and $C > 0$ is a stop-gradient critic scale (Fujimoto and Gu, 2021). Ordinarily, a Polyak copy of this actor enters Eq. (1), while the online actor is deployed. Target-policy movement and deployment reach therefore follow the same branch.

A metric proximal update with step $h > 0$ takes the form

$$\pi^+ \in \arg \min_{\pi} \left\{ \hat{\mathcal{E}}(\pi) + \frac{1}{2h} d_{\mathcal{M}}^2(\pi, \pi^{\text{ref}}) \right\}. \quad (3)$$

This minimizing-movement construction is associated with metric gradient flows (Ambrosio et al., 2005; Otto, 2001). For deterministic state-conditioned policies,

$$W_2^2(\delta_{\mu(s)}, \delta_{\nu(s)}) = \|\mu(s) - \nu(s)\|_2^2. \quad (4)$$

Thus the TD3+BC anchor is a Wasserstein–2 movement cost up to fixed scaling. We distinguish its finite-step variational meaning from the small-step, fixed-critic gradient-flow limit and from large-budget coupled training.

4 BUDGETED ACTOR REFINEMENT

BAR maintains K independently initialized, persistent deterministic actors with separate optimizer states. At each delayed update they take one Adam step in chain order on the same minibatch; Figure 1 gives the exact live ordering. The online critic is fixed within the actor chain; we denote its first Q-function by $\hat{Q}$. We first derive the proximal operator motivating these updates and then state its live-algorithm boundary.

4.1 Offline actor updates as proximal improvement

Let $\rho_{\mathcal{D}}$ be the empirical state distribution and consider a deterministic actor $\mu : \mathcal{S} \rightarrow \mathcal{A} \subset \mathbb{R}^d$. At one training iteration, the critic parameters are held fixed while the actor chain is updated. We write the current critic as $\bar{Q}_t$, use $\epsilon = 10^{-6}$, and define the implementation-level proximal scales by

$$\begin{aligned} C_{1,t}^{\text{P}} &= \text{sg}\left(\mathbb{E}_s|\hat{Q}_t(s, \mu_{1,t}^{\text{pre}}(s))| + \epsilon\right), \\ C_{k,t}^{\text{P}} &= \text{sg}\left(\mathbb{E}_s|\hat{Q}_t(s, \text{sg } \mu_{k-1,t}^+(s))| + \epsilon\right), \quad k \geq 2. \end{aligned} \quad (5)$$

The normalized critic is $\bar{Q}_{k,t}(s, a) = \hat{Q}_t(s, a)/C_{k,t}^{\text{P}}$. Here *pre* and the superscript $+$ denote actions before and after the current actor update, and *sg* denotes stop-gradient. We suppress t and the superscript when discussing the ideal operator. A fixed positive C only rescales gradient-flow time; recomputing C_k changes the finite-step schedule. Our finite-step statements condition on the realized scales.

For deterministic policies and a state distribution ρ , define the action-averaged Wasserstein metric

$$\mathbf{d}_{d,\rho}^2(\mu, \nu) := \mathbb{E}_{s \sim \rho} \left[\frac{1}{d} \|\mu(s) - \nu(s)\|_2^2 \right]. \quad (6)$$

The squared quantity is d^{-1} times the state-averaged squared W_2 distance between state-conditioned Dirac policies. Its square root $\mathbf{d}_{d,\rho}$ is a pseudometric on pointwise policy maps and a metric after identifying maps equal ρ -almost everywhere. We abbreviate $\mathbf{d}_d = \mathbf{d}_{d,\rho_{\mathcal{D}}}$ on the actor-training state distribution. This is an $L_2(\rho)$ geometry on conditional Dirac policy maps with the state marginal held fixed, not Wasserstein transport between policy-induced state-action occupancy measures. The population counterpart of squared behavior cloning also admits a deterministic reference. Let P_B be the data-generating behavior distribution, let ρ_B be its state marginal, and define $b(s) = \mathbb{E}_{P_B}[a | s]$. Then

$$\mathbb{E}_{(s,a) \sim P_B} \frac{1}{d} \|\mu(s) - a\|_2^2 = \mathbf{d}_{d,\rho_B}^2(\mu, b) + \text{const}, \quad (7)$$

where the constant is the conditional action variance and does not depend on μ . The empirical actor loss is a Monte Carlo estimate of the left-hand side, so samplewise squared anchoring targets the conditional-mean behavior map at the population level.

Given a reference policy ν , step $h > 0$, and fixed normalized critic $\bar{Q}$, define the proximal refinement operator by

$$\mathcal{P}_h^{\bar{Q}}(\nu) \in \arg \min_{\mu \in \mathcal{M}} \left\{ -\mathbb{E}_s \bar{Q}(s, \mu(s)) + \frac{1}{2h} \mathbf{d}_d^2(\mu, \nu) \right\}. \quad (8)$$

At hop k , setting $\bar{Q} = \bar{Q}_k$ and multiplying by $2h$ gives the practical actor objective

$$\mathcal{L}_{h,k}(\mu | \nu) = -2h \mathbb{E}_s \bar{Q}_k(s, \mu(s)) + \mathbf{d}_d^2(\mu, \nu). \quad (9)$$

Proposition 1 (Behavior anchoring is Wasserstein proximal improvement). *For deterministic policies, the normalized anchored objective*

$$-\alpha \mathbb{E}_s \bar{Q}(s, \mu(s)) + \mathbf{d}_d^2(\mu, \nu)$$

is Eq. (8), up to multiplication by a positive constant, with $h = \alpha/2$. Equivalently, the raw-critic objective $-\lambda \mathbb{E}_s \hat{Q}(s, \mu(s)) + \mathbf{d}_d^2(\mu, \nu)$ has $h = \lambda C/2$ when C is fixed. The population counterpart of squared dataset anchoring uses $\nu = b$ by Eq. (7).

The proposition identifies an update operator relative to the *estimated* critic; true-return improvement lies outside the scope of this certificate.

4.2 Re-centered composition and actor-critic dataflow

For population analysis, let $\mu_0 := b$ denote the conditional-mean behavior map and let $h_{1:K}$ be positive step sizes. The idealized BAR chain is

$$\mu_k \in \mathcal{P}_{h_k}^{\bar{Q}_k}(\text{sg } \mu_{k-1}), \quad k = 1, \dots, K. \quad (10)$$

The implementation approximates each subproblem with one Adam update. Its first actor μ_1 uses sampled dataset actions as its reference; later actors $\mu_2, \dots, \mu_K$ use the preceding online actor under the same current critic. Only the Polyak copy of μ_1 enters the TD target, while μ_K supplies the evaluated action. Because the later persistent actors are independently initialized, their anchoring gradient need not vanish even at $h = 0$ when predecessor and current actor differ. The ideal proximal map therefore motivates, but does not certify, the live one-Adam-step chain.

Our fixed-budget schedule uses the implementation parameter $T = \tau = \alpha/2$, sets $h_k = T/K$, and holds T fixed. Thus $K = 1$ recovers the TD3+BC coefficient $\alpha = 2T$; T is neither a compute budget nor

One BAR training iteration. The K actors are persistent, independently initialized, and have separate Adam states. 1. Form the Bellman target with μ_1^- and update the online critic. 2. On a delayed step, update μ_1 once toward sampled dataset actions. 3. Polyak-update μ_1^- and the target critics with rate .005. 4. For $k = 2:K$, use the updated sg μ_{k-1}^+ as reference and update μ_k once on the same mini-batch, holding the online critic fixed. 5. Use μ_1^- in future Bellman targets and deploy μ_K.	
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Figure 1: Actual update and routing order. Each hop is one optimizer step, not an exact proximal solve.

a realized distance constraint. Increasing K jointly changes local coefficient, re-centering frequency, actor count and compute, normalization trajectory, and target/deployment dataflow. Each K is therefore a complete randomized procedure, and its seed average estimates marginal end-to-end performance. Identical critic initialization across depths is a common-random-number variance-reduction design, not a validity condition.

On the next-state distribution $\rho'_{\mathcal{D}}$ used by the TD target, define the action-averaged population displacements

$$\begin{aligned} D_{\text{critic}} &:= d_{d, \rho'_{\mathcal{D}}}^2(\mu_1, \mu_0), \\ D_{\text{final}} &:= d_{d, \rho'_{\mathcal{D}}}^2(\mu_K, \mu_0). \end{aligned} \quad (11)$$

A one-step actor has $D_{\text{critic}} = D_{\text{final}}$; BAR assigns the two population quantities to distinct policy branches. On fixed validation batches, we measure the sample-anchored online branches and the deterministic next-state Polyak target branch:

$$\begin{aligned} \hat{D}_{\text{critic}} &= \mathbb{E}_i \|\mu_1(s_i) - a_i\|_2^2/d, \\ \hat{D}_{\text{final}} &= \mathbb{E}_i \|\mu_K(s_i) - a_i\|_2^2/d, \\ \hat{D}_{\text{target}} &= \mathbb{E}_i \|\mu_1^-(s'_i) - a_{i+1}\|_2^2/d. \end{aligned} \quad (12)$$

The direct audit retains only bootstrapped transition pairs; a smoothed version uses common TD3 target-noise draws. Sample anchoring adds a common conditional-action-variance floor in population differences. These are movement measurements, not critic-reliability estimates.

4.3 Finite-step statements

For a re-centered hop where μ_{k-1} is feasible, suppose the realized actor satisfies $\mathcal{L}_{h_k, k}(\mu_k \mid \mu_{k-1}) \leq \mathcal{L}_{h_k, k}(\mu_{k-1} \mid \mu_{k-1}) + \varepsilon_k$. Rearranging the practical

objective gives the conditional estimated-critic margin

$$\begin{aligned} m_k^{\hat{Q}} &:= \frac{2h_k}{C_k} \mathbb{E}_s [\hat{Q}(s, \mu_k(s)) - \hat{Q}(s, \mu_{k-1}(s))] \\ &\quad - d_d^2(\mu_k, \mu_{k-1}) \geq -\varepsilon_k. \end{aligned} \quad (13)$$

For $\hat{Q} = Q^\dagger + e$, this transfers to a trusted objective as

$$m_k^{Q^\dagger} = m_k^{\hat{Q}} - \frac{2h_k}{C_k} \mathbb{E}_s [e(s, \mu_k(s)) - e(s, \mu_{k-1}(s))]. \quad (14)$$

The first sample-anchored hop lacks a feasible deterministic dataset-action comparator, and one Adam step supplies no bound on the unmeasured ε_k . These statements therefore diagnose a realized learned objective; they guarantee neither exact proximal solution nor monotone return. The Supplement gives the derivation and bounded-error corollary.

Proposition 2 (First-hop TD-target perturbation bound). *For clarity, consider a noiseless TD target using μ_1 , $y_{\mu_1} = r + \gamma \min_j Q_j^-(s', \mu_1(s'))$, and suppose each target critic $Q_j^-(s', \cdot)$ is $L(s')$ -Lipschitz. For any reference μ_0 ,*

$$\begin{aligned} \|y_{\mu_1} - y_{\mu_0}\|_{L_2(\mathcal{D})} &\leq \gamma \left(\mathbb{E}_{s'} L(s')^2 \right. \\ &\quad \left. \cdot \|\mu_1(s') - \mu_0(s')\|_2^2 \right)^{1/2}. \end{aligned} \quad (15)$$

If $L(s') \leq \bar{L}$, the right-hand side is at most $\gamma \bar{L} \sqrt{d D_{\text{critic}}}$. Conditional on μ_1 , later actors do not enter this one-step target perturbation.

Proposition 2 bounds a one-step value-target difference, not the stability of the learned critic. Its architectural content is that later refinement can alter D_{final} without directly entering this bound. With a Polyak target network, μ_1 is replaced by its target copy. A same-noise comparison under TD3 target smoothing obeys the same bound because action clipping is non-expansive. The current-state $\hat{D}_{\text{critic}}$ omits Polyak lag and target noise, whereas $\hat{D}_{\text{target}}$ measures the Polyak actor directly and has a smoothed counterpart. Neither displacement measures the action Lipschitz factor or the resulting change in critic value.

The separation introduces a complementary evaluation-policy mismatch: critic targets follow the μ_1 branch, whereas evaluation follows μ_K . Increasing D_{final} at fixed D_{critic} can therefore make the critic less representative of the evaluated policy. We interpret BAR as exposing a measurable target/deployment trade-off; its empirical return captures the net effect of target movement, evaluation mismatch, actor compute, and the coupled optimization trajectory.

4.4 Small-budget flow regime and its boundary

With a fixed critic and common positive scale C , inactive projection, and an exact statewise solve, linearized and ideal proximal steps are βg and $\beta g + \beta^2 Hg + O(\beta^3)$. Repeated composition of one consistent near-identity map has fixed- K , $T \rightarrow 0$ flow bias $O(T^2/K)$ (Supplement), but persistent actors need not satisfy that premise. This limit does not explain the coupled large- T chain. For a ReLU critic, the explicit update and same-region stationary proximal branch coincide before the first activation or box boundary. A pre-locked Hopper/Walker audit finds task-equal residence .954, .911, .842, .748 over 6144 eligible anchors at probe horizons .025, .05, .1, .2; the .2 task range is .310–.980. This bounds the local identity without claiming return effects, a global certificate, or a learned $1/K$ slope.

5 EXPERIMENTS

We evaluate BAR on the nine D4RL locomotion tasks formed by HalfCheetah, Hopper, and Walker2d with medium, medium-replay, and expert datasets (Fu et al., 2020). These tasks directly exercise deterministic continuous control and provide distinct behavior-data regimes for a target-routing stress test. Base implementations follow CORL configurations (Tarasov et al., 2022); scores are D4RL normalized returns. Additional per-task results and diagnostic definitions are in the Supplementary Material.

Fixed-budget protocol and estimand. We hold a total nominal budget T fixed, split it across K actor hops as $h = T/K$, and sweep

$$T \in \{0.05, 0.1, 0.2, 0.4, 0.7, 1.5, 2.5, 4, 7, 10, 12, 14, 17, 20\}.$$

over all nine tasks. Both grids use seeds 0–3, for $9 \times 14 \times 4 = 504$ runs per depth. Proximal seeds 0–1 and 2–3 were executed on separate hosts and form disjoint replication pairs. The grid is a high-resolution stability phase diagram rather than a final benchmark alone. For $K = 1$, Proposition 1 maps T to the normalized TD3+BC coefficient $\alpha = 2T$.

We compare TD3+BC ($K = 1$), proximal BAR with $K \in \{2, 3, 4\}$ (BAR-Prox), and an action-metric-matched projected linearized target with $K \in \{2, 3\}$ (BAR-Lin):

$$a_k^{\text{tar}} = \Pi_{\mathcal{A}} \left[\nu_k + d \frac{T}{K} \frac{\nabla_a \widehat{Q}(s, \nu_k)}{C_{k,t}^L} \right], \quad (16)$$

$$\nu_1 = a, \quad \nu_k = \text{sg } \mu_{k-1,t}^+(s) \quad (k \geq 2).$$

At each hop, BAR-Lin detaches the reference, recomputes the $\widehat{Q}$ gradient and scale, clips the target to the action box, and takes one mean-squared-error optimizer step with the corresponding persistent actor. Its first-hop scale is

$$C_{1,t}^L = \text{sg} \left(\mathbb{E}_{(s,a) \sim \mathcal{D}} |\widehat{Q}_t(s, a)| + \epsilon \right),$$

whereas $C_{k,t}^L = C_{k,t}^P$ for $k \geq 2$. Thus the factor d matches the action metric, but the first-hop normalization point differs from BAR-Prox; the comparison does not match the complete scale schedule. It uses the same first-hop target and final-hop evaluation routing. All conditions match datasets, critic updates, and the 10^6 -update horizon; every headline score is the ten-episode D4RL normalized mean at the final checkpoint. Intermediate evaluation cadence varies by run family and does not enter this estimand. A K -hop condition performs K actor optimizer steps per delayed update, so depth comparisons are between complete training procedures. The end-to-end intervention changes local coefficient, re-centering frequency, actor compute, normalization trajectory, routing, and optimization history.

We test whether this intervention reduces large-budget low-return outcomes, persists under the linearized realization, separates first-hop exposure from final-policy reach, and exposes critic-validity signatures of failure.

5.1 The advantage emerges beyond the local regime

Figure 2 and Table 1 reveal three empirical regions. On $T = .05, .1, .2$, which overlaps the independently probed local horizons, depth gives no consistent return ordering. TD3+BC peaks at $T = 1.5$ and then leaves the high-return plateau: its mean is 67.41, 45.93, 30.36 at $T = 2.5, 4, 7$, whereas P4 gives 81.25, 82.06, 81.79. At $T \geq 10$ every method eventually degrades, but the proximal tail remains depth ordered. These are descriptive regime labels, not predeclared bins. The residence audit and the $T = 20$ projection diagnostic bracket, but do not directly locate, where the local assumptions fail.

The high-budget estimand averages seeds and the seven $T \geq 4$ budgets within each task, then weights tasks equally. A seed-mean environment–budget cell is descriptively collapsed below score 20. Across proximal $K = 1, 2, 3, 4$, the mean rises $25.38 \rightarrow 41.20 \rightarrow 51.92 \rightarrow 63.97$ while collapsed cells fall $34 \rightarrow 22 \rightarrow 16 \rightarrow 8$ of 63. $K = 4$ gains 12.06 over $K = 3$ with task-resampling interval [3.84, 21.11] and seven of nine positive task means, yet its cellwise median is only +.34: eight rescues, no reversals, contribute 54.7% of the

Table 1: Aggregate fixed-budget results. Every entry uses four seeds. Best (T) is the descriptive maximum over the shared 14-value grid; High denotes $T \geq 4$. Collapse entries are seed-mean cells / raw seeded runs.

Method	$T = 1.5$	Best (T)	High	Collapse
TD3+BC	74.79	74.79 (1.5)	25.38	34 / 142
BAR-P2	78.05	80.66 (2.5)	41.20	22 / 111
BAR-P3	78.95	81.41 (2.5)	51.92	16 / 84
BAR-P4	77.70	82.06 (4)	63.97	8 / 54
BAR-L2	75.13	78.43 (2.5)	38.43	25 / 112
BAR-L3	77.74	78.20 (2.5)	46.44	21 / 95

gain. The $T \geq 10$ means are 20.27, 31.01, 42.23, 56.79. BAR-Lin $K = 3$ raises the high-budget mean from 38.43 to 46.44 and reduces collapsed cells from 25 to 21; its +1.37 cellwise median indicates a broader shift. Multi-hop thus shifts the transition chiefly by rescuing the lower tail, not by uniformly lifting every cell.

The high-budget aggregate is a robustness estimand, not a deployment rule. At TD3+BC’s grid-best shared budget, $T = 1.5$, BAR-P3 and P4 score 78.95 and 77.70 versus 74.79. Allowing one globally shared grid value per method, BAR-P4 reaches 82.06 at $T = 4$. In a post-hoc tuning-transfer sensitivity that selects each method’s T on two dynamics families and tests on the third, P4 averages 78.44 versus 74.79 for TD3+BC (+3.65; 7/9 task differences positive). Family gains vary, so these summaries demonstrate a practical regime but do not provide an offline selection rule.

Task dependence and replication. Responses range from smooth to sharply nonmonotone, so there is no universal critic-valid radius. Nominal T is the strongest absolute within-environment rank predictor of return in eight of nine environments; realized movement is a mechanism readout, not a substitute for the intervention. P4–P3 is positive within all three dynamics families (+.79, +13.04, +22.33), and every leave-one-family-out mean remains positive (+6.92 to +17.69). Both host-separated seed pairs reproduce the high-budget ordering: 25.97, 42.02, 52.78, 63.61 and 24.79, 40.38, 51.05, 64.33. These are disjoint-seed replications of one implementation.

The linearized sweep shows that the gain is not unique to the proximal loss. At high budgets, BAR-Lin with $K = 3$ gains 8.01 points over $K = 2$ (95% task-resampling interval [3.56, 13.53]; eight of nine positive task means) and reduces collapse cells from 25 to 21. Both disjoint seed pairs reproduce the direction, with gains of 7.23 and 8.80. BAR-Prox remains stronger overall, so these are comparisons of complete training procedures rather than isolated integrator effects.

5.2 Transition and large-budget failure are nonlocal

At $T = 20$, the projected BAR-Lin $K = 2$ target is clipped at an action bound in 94–95% of states, and the effective action-step coefficient h_{eff} is only 14% and 9% of the nominal dT/K coefficient in the two hops. These are medians over 18 environment–seed runs per hop. The practical operation is projected target generation followed by amortized neural regression under a moving critic. The sweep measures the resulting end-to-end stability envelope; local truncation error lies outside this estimand.

Figure 3 shows the common failure signature. Environment return supplies the external label: collapsed runs score below 20. These runs have orders-of-magnitude larger TD-error tails, larger critic magnitudes, and larger realized movements. The controlled common critic is the frozen final TD3+BC $T = 1$ critic matched by environment and seed. A run-total gain sums the mean Q change across actor hops on the shared state batch. For collapsed BAR-Lin runs, its median under the jointly trained critic is 3.33×10^{11} ; under the common critic it is -1.42 , compared with +2.11 for stable runs. For BAR-Prox, the common-critic gain changes from +2.51 in stable runs to -1.06 in collapsed runs. This sign reversal diagnoses cross-critic disagreement consistent with Eq. (14). A separate four-environment seed-2/3 sensitivity audit, using a fixed TD3+BC $T = .05$ reference critic, preserves the split: all 45 stable gains are positive and all 35 collapsed gains are negative (Supplement).

A simulator-reference audit provides targeted evidence of critic miscalibration in the screened checkpoints: the median signed error $\hat{Q}_1(s, a_K) - Q_{\text{MC}}^{\rho_1}(s, a_K)$ is $+1.70 \times 10^{12}$ in five collapsed checkpoints and -41.4 in 55 stable ones, using the median of checkpoint-level state medians. Episode-cluster intervals exclude zero for critic error at 58 of 60 checkpoints, and restoring the sampled episode clock preserves the error sign at all 60. Its action-ranking subset is sparse, so we use it only as targeted checkpoint-level evidence.

5.3 How multi-hop helps: exposure versus reach

The phase diagram points to an exposure–reach mechanism rather than a small-step accuracy effect. For fixed T , only BAR’s first T/K branch enters the Bellman target, while the later branches can extend the deployed policy.

On identical next-state batches, P3/P4 first-actor displacement is below TD3+BC in 88/90 and 89/90 cells (median squared-displacement ratios .650, .565), while

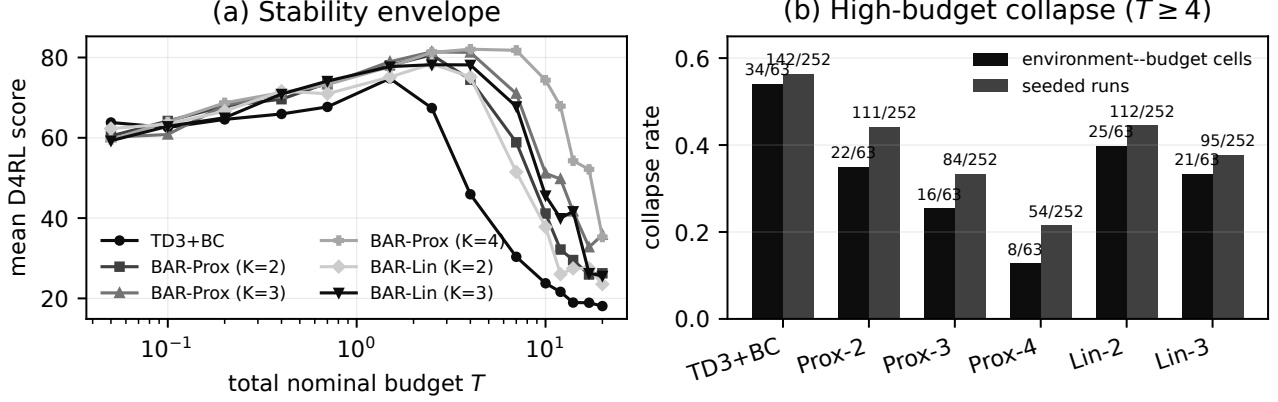


Figure 2: Fixed-budget sweep over nine locomotion tasks. (a) Mean score over tasks. (b) Collapse rates for $T \geq 4$, reported both by environment-budget cell and by seeded run. All curves and cell means use four seeds. Both realizations broaden the stable high-budget region with depth.

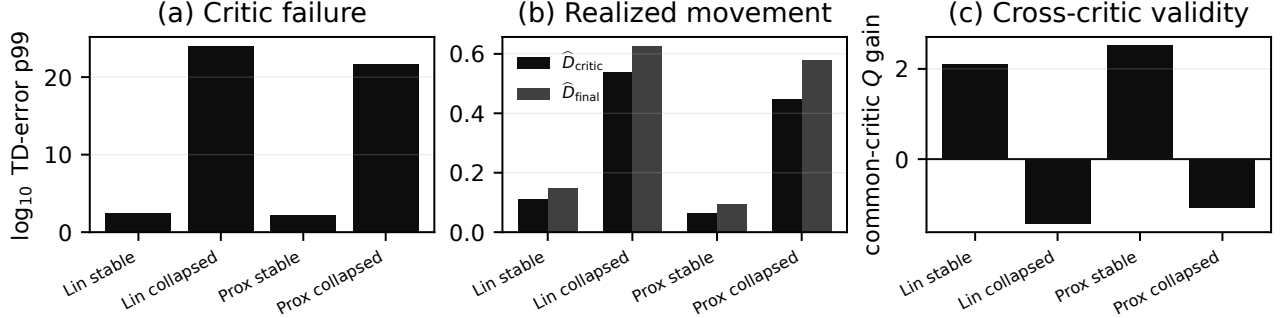


Figure 3: Median diagnostics over the matched $T \in \{4, 7, 10, 14, 20\}$ audit. Proximal and linearized realizations share a diagnostic pattern consistent with critic miscalibration: critic errors and realized movements increase sharply, and run-total gains under a common critic become negative. Environment return labels failure; the common critic diagnoses cross-critic disagreement.

final-actor displacement is below it in only 37/90 and 44/90 (1.060, 1.025). In every P3 and P4 cell the final actor lies farther from the paired next dataset action than the first Polyak actor (median RMS ratios 1.30, 1.36). Under one matched TD3+BC target critic and identical smoothing, the operational TD-target perturbation is also lower in 86/90 and 88/90 cells (.717, .617), with all nine task medians below one. These learned-critic readouts establish asymmetric movement and reduced critic-space perturbation, not critic accuracy or return causality. Seed-2/3 and lower-exposure return counterexamples are reported in the Supplement.

BAR-Prox also realizes a smaller first-hop proxy than BAR-Lin once $T \geq 7$. Exactly one realization collapses in 11/90 matched cells; in every case the stable run has smaller first and final movement. This associates controlled movement with stability without isolating a causal route.

Table 2: P3 policy-separation control over 90 matched task-budget-seed pairs. Collapse is the descriptive score-20 count before / after seed averaging.

Condition	Mean	Raw < 20	Cell < 20
BAR-P3 rerun	57.67	23/90	9/45
Two-actor control	59.22	21/90	9/45

5.4 Separation explains part of the multi-hop gain

We test sequential re-centering with a two-seed control whose directly data-anchored target/deployed actors use $T/3$ and T , compared only with a contemporaneous BAR-P3 rerun. Control minus BAR is +1.56 with task-resampling interval $[-1.52, 5.44]$, while its paired median is $-.42$ and BAR wins 52/90 pairs; only 2/9 task means favor the control. Thus policy separation

attains the observed P3 aggregate regime without sequential re-centering. This is a bundled, non-compute-matched comparison that also changes anchoring and critic trajectory, not an equivalence or re-centering-only effect. A completed P4 score merge is unresolved but cross-stack, so it fails the locked protocol (Supplement).

Eight-run controls give 71.4, 70.0, 59.4 for compute-matched sequential, fixed-reference, and direct-final refinement; slower target averaging does not rescue $K = 1$. Route shadows miss their common-continuation rule (Supplement).

5.5 Regime synthesis and limitations

The results support one regime-based reading. At small horizons, the Wasserstein/JKO view explains the common local direction, and ReLU residence makes the exact affine identity empirically relevant. It does not predict a return ordering. As T leaves the small-budget plateau, TD3+BC couples a large policy move directly into bootstrapping; BAR reduces that first-branch exposure while retaining final reach, and the stability advantage opens. In the extreme tail, all procedures become nonlocal and degrade, but added depth moves the collapse boundary and rescues its lower tail. Policy separation is sufficient for the P3 aggregate, while the P4 chain increment remains unresolved. The live K intervention still bundles step size, re-centering, compute, normalization, routing, and optimization history.

Both headline score grids use four seeds, and the proximal grid forms two host-separated pairs. The full P3 movement and exposure audit uses seeds 0–1; a restricted four-environment seed-2/3 exposure audit supplies a sensitivity check, while the P4 target dump uses four seeds. For proximal seeds 2–3, provenance consists of the archived launch configuration and score matrices; exact per-run manifests were not retained, and post-hoc P4 manifests do not reconstruct them. The protocol-v2 full P4 final-displacement audit is limited to seeds 0–1. The tasks span three dynamics families, collapse is descriptive, and controls are post hoc. The 90-pair policy-separation control uses seeds 0–1, is not compute matched, and includes ten Walker2d-expert cells completed through a documented CPU recovery chain (eight resumed and two CPU-only). We provide no offline rule for choosing T or K . BAR makes K actor optimizer calls per delayed update and stores K persistent actors, so actor work and persistent actor state scale as $O(K)$. On one H200/JAX 0.10.2 stack, compiled actor-phase time rises from 1.435 to 1.887 ms and scoped backend high-water memory from 9.68 to 18.05 MB between $K = 1$ and $K = 4$ (1.316 $\times$, 1.863 $\times$); these one-stack measurements are not end-to-

end scaling. The ideal fixed-critic analysis motivates but does not certify the live chain.

6 CONCLUSION

BAR turns the JKO motivation into a finite-step stability intervention. The small- T limit explains why ideal refinements share a local flow, but the empirical advantage appears when TD3+BC leaves that regime. Across four seeds, the $T \geq 4$ mean rises from 25.38 at $K = 1$ to 63.97 at $K = 4$ as low-return cells fall from 34/63 to 8/63; both disjoint seed pairs reproduce the ordering. Matched audits reveal the corresponding signature: added depth contracts the target-coupled first branch without typically contracting the deployed policy, while most of the P4 gain comes from lower-tail rescue. At extreme budgets the local theory is inapplicable and every method degrades, so depth shifts rather than removes the failure boundary. Within BAR, K is therefore a useful finite-step procedure control, not merely a discretization-accuracy knob. The two-actor result identifies policy separation as the simpler sufficient mechanism hypothesis; a protocol-admissible P4 control remains needed to isolate the sequential chain’s incremental value.

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Checklist

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes/No/Not Applicable]
Yes. Sections 3–4 define the setting, actor chain, losses, and assumptions.
 - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes/No/Not Applicable]
Yes. Section 4 states that BAR uses K persistent actors and K actor optimizer calls per delayed update; Eq. (13), Proposition 2, and the local-idealization subsection state the theoretical properties and scope. The supplement reports a verified one-stack time and memory profile.
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes/No/Not Applicable]
Partial. The anonymized release contains the BAR launch, evaluation, diagnostic, and verification code plus the compact policy-separation audit. The control-run configs identify local training-source files by SHA-256, but did not record a clean Git revision.
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. [Yes/No/Not Applicable]
Yes. Each proposition states its assumptions, and the local-flow assumptions are separated from the coupled large-budget regime.
 - (b) Complete proofs of all theoretical results. [Yes/No/Not Applicable]
Yes. Complete proofs are provided in the supplementary material.
 - (c) Clear explanations of any assumptions. [Yes/No/Not Applicable]
Yes. Sections 4 and 5 explain the learned-critic, fixed- K , small-step, and implementation restrictions.
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes/No/Not Applicable]
Yes. Released CSVs, figure-generation code, a numerical claim verifier, and build instructions reproduce every main figure and aggregate table.
 - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes/No/Not Applicable]
No. All recoverable settings are listed, but exact per-run BAR-P4 configurations and manifests for proximal seeds 2–3 are unavailable; the policy-separation configs also omit a source revision and environment lock.
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes/No/Not Applicable]
Yes. Captions and text define the equally

task-and-budget-weighted high-budget estimand, seed-mean cells, raw seeded runs, collapse thresholds, paired ratios, and task-resampling intervals. Both headline grids use four seeds; proximal seeds form two host-separated pairs. Mechanism audits use the explicitly reported two- or four-seed subsets.

- (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes/No/Not Applicable]

No. The released artifact fully records the H200 used for the actor-cost profile and the policy-separation CPU recovery chain, but does not contain a complete, independently verified inventory of all training hardware.

- 4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:

- (a) Citations of the creator If your work uses existing assets. [Yes/No/Not Applicable]

Yes. D4RL, TD3+BC, and the CORL implementation basis are cited.

- (b) The license information of the assets, if applicable. [Yes/No/Not Applicable]

Yes. The code release identifies its license; upstream dependencies retain their own licenses.

- (c) New assets either in the supplemental material or as a URL, if applicable. [Yes/No/Not Applicable]

Yes. An anonymized code and compact-results artifact accompanies the submission.

- (d) Information about consent from data providers/curators. [Yes/No/Not Applicable]

Not Applicable. We use public simulated-control benchmark datasets and collect no human data.

- (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Yes/No/Not Applicable]

Not Applicable. The assets contain simulated locomotion trajectories and no personal or sensitive content.

- 5. If you used crowdsourcing or conducted research with human subjects, check if you include:

- (a) The full text of instructions given to participants and screenshots. [Yes/No/Not Applicable]

Not Applicable.

- (b) Descriptions of potential participant risks, with links to Institutional Review Board

(IRB) approvals if applicable. [Yes/No/Not Applicable]

Not Applicable.

- (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Yes/No/Not Applicable]

Not Applicable.